

6.1.4 Lab: Connect a Cable Modem

null

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Task Summary

Required Actions

- ✓ Connect the cable modem to the Internet using the RG-6 cable
- ✓ Connect the computer to the cable modem using the Ethernet (RJ-45) cable
- ✓ Connect the power cable to the cable modem
- ✓ Confirm that the computer is properly connected to the Internet.

Explanation

To complete this lab:

- Add the cable modem/router to the Workspace. Connect the modem to the WAN connection using the RG-6 coaxial cable.
- Connect the computer to the cable modem using a Cat6a Ethernet cable.
- Use the AC to DC power adapter to plug in the modem.

Complete the following steps:

1. On the Inventory, expand the **Networking Devices** category.
2. Drag the cable modem from the Inventory to the Workspace.
3. To connect the modem to the Internet connection, switch to the back view of the cable modem.
4. On the Inventory, expand the **Cables** category.
5. Select the RG-6 coaxial cable.
6. In the Selected Item Connection window, drag a connector to the port on the cable modem.
7. In the Selected Item Connection window, drag the other connector to the port on the wall plate.
8. To connect the computer to the cable modem, switch to the back view of the computer.
9. On the Inventory, select the Cat6a Ethernet cable under the Cables category.
10. In the Selected Item window, drag a connector to the Ethernet port on the cable modem.
11. In the Selected Item window, drag the unconnected end of the Ethernet port to the computer.
12. To plug in the cable modem, select the power adapter on the Inventory.
13. In the Selected Item window, drag the DC power connector to the port on the cable modem.
14. In the Selected Item window, drag the AC adapter end to the power outlet.

Accessible Steps to Complete the Lab:

1. **Add the Cable Modem to the workspace:**
 - a. Press **ALT-I** to access the **Inventory** menu.
 - b. Tab to **Networking Devices** and press **Enter**.
 - c. Tab to the **Cable Modem/Router** and press **Enter**.
 - d. Press **Enter** again to Add the **printer** to the **workbench**.
 - e. On the **Select Location to Add Item** window:
 1. In the drop-down menu under **Connector End to Connect**, press the **Down arrow** key until **Cable Modem/Router** is shown.
 2. Press **Tab** to access the **Destination** drop-down menu.
 3. Use the **Down arrow** key until **Bench** is shown and press **Enter**.
 4. Press **Tab** twice and then **Enter** to **Confirm**.
 - f. The **modem** should now be on the **workbench**.
2. **Connect the modem to the ISP's cable line:**
 - a. Press **ALT-I** to access the **Inventory** menu.
 - b. Tab to **Cables** and press **Enter**.
 - c. Tab to **Coaxial Cable, RG-6 (75 ohms), F Type** and press **Enter**. Press **Enter** again to add the cable.
 - d. On the **Select Location to Add Item** window:
 1. In the drop-down menu under **Connector End to Connect**, press the **Down arrow** key until **Coaxial Type F Connector** is shown.
 2. Press **Tab** to access the **Destination** drop-down menu.
 3. Use the **Down arrow** key until **Cable Port Cable Modem** is shown and press **Enter**.
 4. Press **Tab** twice and then **Enter** to **Confirm**.
 - e. Press **Alt-W** to access the **Workspace**.
 - f. Tab until the arrow next to **Cable Modem/Router** is highlighted and press **Enter**.
 - g. Tab two times so the arrow next to **Coaxial Type F Connector** is highlighted and press **Enter**.
 - h. Tab one time so **Connector, Coaxial, Type F, Male** is highlighted and press **Enter**. Press **Enter** again to add the connector.
 - i. In the **Select Location to Add Item** Window:
 1. Tab to the **Destination** field and press **Enter**. Use the **down arrow** key until **Cable TV Outlet, AC 115 volt x2, RJ45 x2, RJ-11 x2, F-type** is selected and press **Enter**.
 2. Press **Tab** twice and then **Enter** to **confirm**.
3. **Connect the modem to the computer:**
 - a. Press **Alt-I** and Tab to **Cables** and press **Enter**.
 - b. Tab to **Cat6a Cable, RJ45** and press **Enter**. Press **Enter** again to add the cable.
 - c. On the **Select Location to Add Item** window:
 1. In the drop-down menu under **Connector End to Connect**, press the **Down arrow** key until **RJ45 Shielded Connector (568B)** is shown.
 2. Press **Tab** to access the **Destination** drop-down menu.
 3. Use the **Down arrow** key until **LAN Port Cable Modem** is shown and press **Enter**.
 4. Press **Tab** twice and then **Enter** to **Confirm**.
 - d. Press **Alt-W** to access the **Workspace**.
 - e. Tab until the arrow next to **Cable Modem/Router** is highlighted and press **Enter**.
 - f. Tab two times so the arrow next to **RJ45 Shielded Connector (568B)** is highlighted and press **Enter**.
 - g. Tab one time so **Connector, Shielded, RJ45, 568B, Male** is highlighted and press **Enter**. Press **Enter** again to add the connector.
 - h. In the **Select Location to Add Item** Window:
 1. Tab to the **Destination** field and press **Enter**. Use the **down arrow** key until **Network Motherboard, Socket AM3, NIC** is selected and press **Enter**.
 2. Press **Tab** twice and then **Enter** to **confirm**.
4. **Provide power to the modem:**

- a. Press **Alt-I** and then **Tab** to **Cables** and press **Enter**.
- b. **Tab** to **Power Adapter, AC to DC** and press **Enter**. Press **Enter** again to add the cable.
- c. On the **Select Location to Add Item** window:
 1. In the drop-down menu under **Connector End to Connect**, press the **Down arrow** key until **DC Power Connector** is shown.
 2. Press **Tab** to access the **Destination** drop-down menu.
 3. Use the **Down arrow** key until **Power Port Cable Modem** is shown and press **Enter**.
 4. Press **Tab** twice and then **Enter** to **Confirm**.
- d. Press **Alt-W** to access the **Workspace**.
- e. **Tab** until the arrow next to **Cable Modem/Router** is highlighted and press **Enter**.
- f. **Tab** two times so the arrow next to **DC Power Connector** is highlighted and press **Enter**.
- g. **Tab** one time so **Connector, Power, AC Adapter, v2015** is highlighted and press **Enter**. Press **Enter** again to add the connector.
- h. In the **Select Location to Add Item** Window:
 1. **Tab** to the **Destination** field and press **Enter**. Use the **down arrow** key until **AC1 Outlet, AC 115 volt x2, RJ45 x2, RJ-11 x2, F-type** is selected and press **Enter**.
 2. Press **Tab** twice and then **Enter** to **confirm**.

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